D Y PATIL INTERNATIONAL UNIVERSITY
Sector 29, Pradhikaran, Akurdi, Pune - Maharashtra, INDIA 411044
(Establishment by Maharashtra Act No. LXIII of 2017)
School of Computer Science, Engineering & Applications

Course Name	Applied Computational Mathematics
Course Code	CSE1001
Credits	3
Prerequisites	Intermediate Mathematics
L-T-P-C	3-0-0-3

COURSE OBJECTIVES

1. To introduce matrix methods and eigenvalue problems in linear algebra.
2. To develop techniques for analyzing and optimizing multivariable functions.
3. To present methods for solving ordinary differential equations.
4. To build understanding of integral and vector calculus in physical applications.

COURSE SYLLABUS

Modules	**Contact Hours**
Module I: Linear Algebra – I - Matrices and matrix operations, Elementary row operations, Row echelon form, Reduced row echelon form, Gauss-Jordan elimination.	7
Module II: Linear Algebra – II - Rank and inverse of a matrix using row operations, Systems of homogeneous and non-homogeneous linear equations, Matrix equation $Ax = b$, Eigenvalues and eigenvectors.	7
Module III: Differential Calculus - Functions of several variables, Partial derivatives, Chain rule, Critical points, Maxima, Minima, Saddle points, Second derivative test, Lagrange multipliers.	7
Module IV: Differential Equations - Introduction to ordinary differential equations (ODEs), Order and degree of ODEs, First-order separable and linear DEs, Second-order linear homogeneous and nonhomogeneous DEs with constant coefficients, Method of variation of parameters.	7
Module V: Integral and Vector Calculus - Basic integration, Double integrals, Triple integrals, Applications in calculating area and volume, Gradient, Directional derivatives, Divergence, Curl, Physical interpretations.	8
Self Study/Concurrent Assessment Hours	9
Total Contact Hours (36+9)	**45**

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Understand the fundamental concepts of matrices, multivariable calculus, ordinary differential equations, and vector calculus.
CO2	Apply matrix operations, system solutions, and differential equations to resolve standard computational problems.
CO3	Compute partial derivatives, multiple integrals, and vector differential operators to solve physical and spatial engineering problems.
CO4	Analyze systems of linear equations, optimization constraints, and multivariable functions to interpret engineering data.
CO5	Examine complex computing problems by integrating mathematical tools, differential methods, and vector analysis for domain-specific solutions.

CO-PO Mapping:			3: Substantial				2: Moderate					1: Slight		
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	1	–	–	–	–	–	–	1	–	–	–	1	1
CO2	3	2	–	1	–	–	–	–	1	–	–	–	1	2
CO3	3	2	–	2	–	–	–	–	1	–	–	–	1	2
CO4	3	3	–	2	–	–	–	–	1	–	–	–	1	3
CO5	3	3	1	3	–	–	–	–	1	–	–	–	2	3

TEXTBOOKS

1. Kreyszig, E. (2011). *Advanced Engineering Mathematics*. 10th ed. Wiley-India.
2. Grewal, B. S. (2014). *Higher Engineering Mathematics*. 43rd ed. Khanna Publishers.
3. Strang, G. (2016). *Introduction to Linear Algebra*. 5th ed. Wellesley-Cambridge Press.
4. Thomas, G. B., Weir, M. D., & Hass, J. (2014). *Thomas' Calculus: Early Transcendentals*. 13th ed. Pearson.

EVALUATION SCHEME

Continuous Assessment	Mid-term	End-term	Total
25	25	50	100